

Egyptian Legal AI

Test & Evaluation Report

Comprehensive Quality Assurance & Performance Analysis

RAG System | E2E Testing | AI Quality | Experiment Tracking

Date: June 23, 2026

Project: Egyptian Labour Law AI Assistant

Law Reference: Law 14 of 2025

LLM: Google Gemini 2.5 Flash

Embedding: BAAI/bge-m3 (1024-dim)

Vector Store: FAISS

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1. Executive Summary

This report presents the comprehensive testing and evaluation results for the Egyptian Legal AI system — an intelligent Retrieval-Augmented Generation (RAG) assistant specialised in Egyptian Labour Law (Law 14 of 2025). The system was evaluated across three complementary dimensions:

- End-to-End (E2E) Testing via Playwright: 13 automated browser tests covering authentication, file management, API health, responsive design, and AI legal accuracy.
- RAG Quality Evaluation via Deepchecks: 15 legal questions spanning 7 topic areas at 3 difficulty levels, measuring keyword coverage, response quality, and legal reference accuracy.
- Experiment Tracking via MLflow: Systematic logging of model parameters, quality metrics, and performance data for reproducibility and comparison.

100%

E2E Pass Rate

100%

RAG Pass Rate

94.5%

Avg Quality Score

28

Total Tests Run

VERDICT: All 28 tests passed successfully — 100% pass rate across all evaluation suites.

2. System Architecture & Technology Stack

The Egyptian Legal AI system is built on a modern full-stack architecture combining a React frontend, FastAPI backend, and a RAG pipeline powered by Google Gemini and FAISS.

2.1 Technology Stack

Component	Technology	Details
LLM	Google Gemini 2.5 Flash	Low latency, high quality reasoning
Embedding Model	BAAI/bge-m3	1024-dimensional multilingual embeddings
Vector Store	FAISS	1,201 indexed chunks from 894 documents
Agent Framework	LangChain ReAct	6 max iterations, tool-based reasoning
Backend	FastAPI + Uvicorn	RESTful API with CORS support
Frontend	React.js	Responsive Arabic-first UI
Database	SQL Server / SQLite	User auth, chat history, messages
OCR	Tesseract + Gemini Vision	Document and image text extraction

2.2 RAG Pipeline Configuration

Chunk Size:	2,000 characters
Chunk Overlap:	200 characters
Dynamic Retrieval:	Enabled (2-6 documents per query)
Query Expansion:	Domain-specific legal term mapping
Document Deduplication:	Content-hash based (first 200 chars)
Response Caching:	In-memory with file-fingerprint invalidation
Max Agent Iterations:	6 (balances depth vs. API rate limits)

3. Testing Methodology

The evaluation framework employs a three-tier testing strategy to validate system correctness, AI quality, and reproducibility. Each tier addresses a distinct risk surface.

Tier 1: End-to-End (E2E) Testing — Playwright

Validates the full application stack from the browser perspective. Tests cover user authentication flows, API endpoint correctness, frontend rendering, responsive design across breakpoints, file upload/download, and AI response validation. All tests run in headless Chromium with Arabic locale (ar-EG) configured.

Coverage: 13 tests

Tier 2: RAG Quality Evaluation — Deepchecks

Evaluates the retrieval and generation quality of the AI agent using 15 curated legal questions across 7 topic areas (leaves, wages, contracts, dismissal, insurance, disputes, safety) at 3 difficulty levels (simple, medium, complex). Each response is scored on 5 quality dimensions with a weighted aggregate.

Coverage: 15 test questions

Tier 3: Experiment Tracking — MLflow

Provides systematic tracking of model configuration parameters, per-question quality metrics, and performance data. Supports historical comparison across evaluation runs and generates persistent artifacts for auditing.

Coverage: Continuous tracking

3.1 Quality Scoring Formula (Deepchecks)

Each RAG response is evaluated using a weighted composite score:

$$\begin{aligned} \text{Overall Score} = & 0.30 \times \text{Keyword Coverage} + 0.20 \times \text{Completeness} \\ & + 0.20 \times \text{Legal References} + 0.15 \times \text{Arabic Quality} + 0.15 \times \text{No Error} \end{aligned}$$

Pass threshold: Overall Score $\geq$ 50%

4. Playwright E2E Test Results

The Playwright E2E suite executed 13 tests in 46.9 seconds using headless Chromium. All tests passed with a 100% success rate.

13	13	0	46.9s
Total Tests	Passed	Failed	Duration

4.1 Detailed Test Results

#	Test Name	Status	Duration	Category
1	API Health Check	PASS	329ms	Infrastructure
2	Login Page Load	PASS	2.9s	UI / Frontend
3	Signup Flow	PASS	5.2s	Authentication
4	Login Flow	PASS	6ms	Authentication
5	Invalid Login Rejection	PASS	6ms	Security
6	File Upload API	PASS	9ms	File Management
7	Files List API	PASS	5ms	File Management
8	API Docs Accessible	PASS	4.6s	Documentation
9	Frontend Responsive Design	PASS	4.1s	UI / Frontend
10	AI Legal Accuracy (4 questions)	PASS	13.5s	AI Quality
11	AI Out-of-Scope Rejection	PASS	1.6s	AI Quality
12	AI Response Consistency	PASS	6.6s	AI Quality
13	AI Arabic Language Quality	PASS	1.3s	AI Quality

4.2 Test Coverage by Category

Category	Tests	Pass Rate	Status
AI Quality	4	100%	ALL PASSED
Authentication	2	100%	ALL PASSED
Documentation	1	100%	ALL PASSED
File Management	2	100%	ALL PASSED
Infrastructure	1	100%	ALL PASSED
Security	1	100%	ALL PASSED
UI / Frontend	2	100%	ALL PASSED

5. Deepchecks RAG Evaluation Results

The Deepchecks evaluation assessed RAG quality across 15 legal questions covering 7 topic areas of Egyptian Labour Law. All questions passed with an average quality score of 94.5%.

15	100%	94.5%	0
Questions	Pass Rate	Avg Score	Failed

5.1 Performance by Difficulty Level

Difficulty	Count	Avg Score	Passed	Status
Simple	5	90.4%	5/5	ALL PASSED
Medium	6	96.0%	6/6	ALL PASSED
Complex	4	97.5%	4/4	ALL PASSED

5.2 Average Quality Dimensions

Metric	Score	Description
Keyword Coverage	88.9%	Required keywords present in response
Completeness	96.0%	Response length and substantiveness
Structure & Formatting	93.3%	Use of bullets, numbers, bold formatting
Legal References	93.3%	Mentions specific articles and law numbers
Arabic Language Quality	100.0%	Percentage of Arabic characters (>30%)
Error-Free Responses	100.0%	Response does not contain error messages

6. MLflow Experiment Tracking

MLflow provides systematic experiment tracking for the evaluation pipeline. Each evaluation run logs model configuration parameters, per-question quality metrics, and generated artifacts for historical comparison and reproducibility.

6.1 Tracked Parameters

Parameter	Value
model_name	models/gemini-2.5-flash
temperature	0.1
max_tokens	4096
top_p	0.95
top_k_documents	6
chunk_size	2000
chunk_overlap	200
embedding_model	BAAI/bge-m3
max_iterations	6

6.2 Tracked Metrics (Latest Run)

Metric	Value	Interpretation
total_tests	15	Full question set
evaluated	15	All questions evaluated
passed	15	All above 50% threshold
failed	0	No failures
pass_rate	100%	Perfect pass rate
avg_overall_score	0.9453	High overall quality
min_score	0.56	Lowest: minimum wage (Q03)
max_score	1.00	12 questions scored 100%
avg_completeness	0.96	Responses are substantial
avg_structure	0.93	Good formatting present
avg_legal_references	0.93	Cites law articles
avg_arabic_quality	1.00	Perfect Arabic output
avg_no_error	1.00	Zero error messages
avg_keyword_coverage	0.89	High keyword hit rate
simple_avg_score	0.904	Simple questions: strong
medium_avg_score	0.960	Medium questions: excellent
complex_avg_score	0.975	Complex questions: highest

7. Per-Question Detailed Breakdown

The following table shows the detailed evaluation results for each of the 15 legal questions in the Deepchecks RAG evaluation suite.

ID	Difficulty	Keywords	Complete	Structure	Legal Ref	Arabic	Pass
Q01	Simple	100%	100%	100%	100%	100%	Y
Q02	Medium	67%	100%	100%	100%	100%	Y
Q03	Simple	0%	80%	67%	50%	100%	Y
Q04	Complex	67%	100%	100%	100%	100%	Y
Q05	Simple	100%	80%	100%	100%	100%	Y
Q06	Medium	100%	100%	100%	100%	100%	Y
Q07	Medium	100%	100%	100%	100%	100%	Y
Q08	Complex	100%	100%	100%	100%	100%	Y
Q09	Medium	100%	80%	33%	50%	100%	Y
Q10	Complex	100%	100%	100%	100%	100%	Y
Q11	Medium	100%	100%	100%	100%	100%	Y
Q12	Simple	100%	100%	100%	100%	100%	Y
Q13	Complex	100%	100%	100%	100%	100%	Y
Q14	Medium	100%	100%	100%	100%	100%	Y
Q15	Simple	100%	100%	100%	100%	100%	Y

7.1 Question Details & Scores

[Q01]	Simple الإجازات	Score: 100%	ما هي مدة الإجازة السنوية للعامل؟
[Q02]	Medium حقوق المرأة العاملة	Score: 90%	ما هي حقوق المرأة العاملة في قانون العمل المصري؟
[Q03]	Simple الأجور	Score: 56%	ما هو الحد الأدنى للأجور؟
[Q04]	Complex إنهاء علاقة العمل	Score: 90%	ما هي إجراءات الفصل التعسفي؟
[Q05]	Simple ساعات العمل	Score: 96%	كم عدد ساعات العمل اليومية؟
[Q06]	Medium عقود العمل	Score: 100%	ما هي شروط عقد العمل؟
[Q07]	Medium الإجازات	Score: 100%	ما هي أنواع الإجازات المتاحة للعامل؟
[Q08]	Complex إنهاء علاقة العمل	Score: 100%	ما هي حقوق العامل عند إنهاء العقد؟
[Q09]	Medium التأمينات الاجتماعية	Score: 86%	ما هي قواعد التأمينات الاجتماعية للعمال؟

[Q10]	Complex النزاعات العمالية	Score: 100%	ما هي إجراءات تسوية النزاعات العمالية؟
[Q11]	Medium السلامة المهنية	Score: 100%	ما هي قواعد السلامة والصحة المهنية؟
[Q12]	Simple الإجازات	Score: 100%	ما هي حقوق العامل في الإجازة المرضية؟
[Q13]	Complex عقود العمل	Score: 100%	ما الفرق بين عقد العمل محدد المدة وغير محدد المدة؟
[Q14]	Medium تشغيل الأطفال	Score: 100%	ما هي قواعد تشغيل الأطفال؟
[Q15]	Simple حقوق المرأة العاملة	Score: 100%	ما هي مدة إجازة الوضع للمرأة العاملة؟

8. Conclusions & Recommendations

8.1 Key Findings

+ 100% E2E Pass Rate

All 13 Playwright tests pass consistently, validating authentication, file management, responsive design, and AI quality.

+ 100% RAG Pass Rate

All 15 legal questions pass the quality threshold. The system correctly handles simple, medium, and complex queries across 7 legal domains.

+ 94.5% Average Quality Score

Responses are well-structured, cite specific legal articles, and maintain high Arabic language quality. Complex queries score highest (97.5%).

+ Robust Error Handling

The parsing error handler successfully recovers from LLM formatting issues, extracting clean answers from malformed ReAct output.

+ Effective Query Expansion

Domain-specific terminology mapping bridges the gap between colloquial legal terms and the precise language of Law 14 of 2025.

8.2 Areas for Improvement

! Minimum Wage Query (Q03 — Score: 56%)

The law does not specify an exact minimum wage figure, leading to lower keyword coverage. Consider adding supplementary data sources (ministerial decrees) to the knowledge base.

! Social Insurance (Q09 — Score: 86%)

Response structure could be improved with bullet points and article references. The insurance law cross-references could be expanded.

! Live vs. Cached Testing

Current cached-mode evaluations ensure consistency, but periodic live API evaluations should be run to detect model drift or API changes.

8.3 Final Verdict

SYSTEM STATUS: PRODUCTION READY

28/28 tests passed | 94.5% average quality | 100% pass rate across all suites